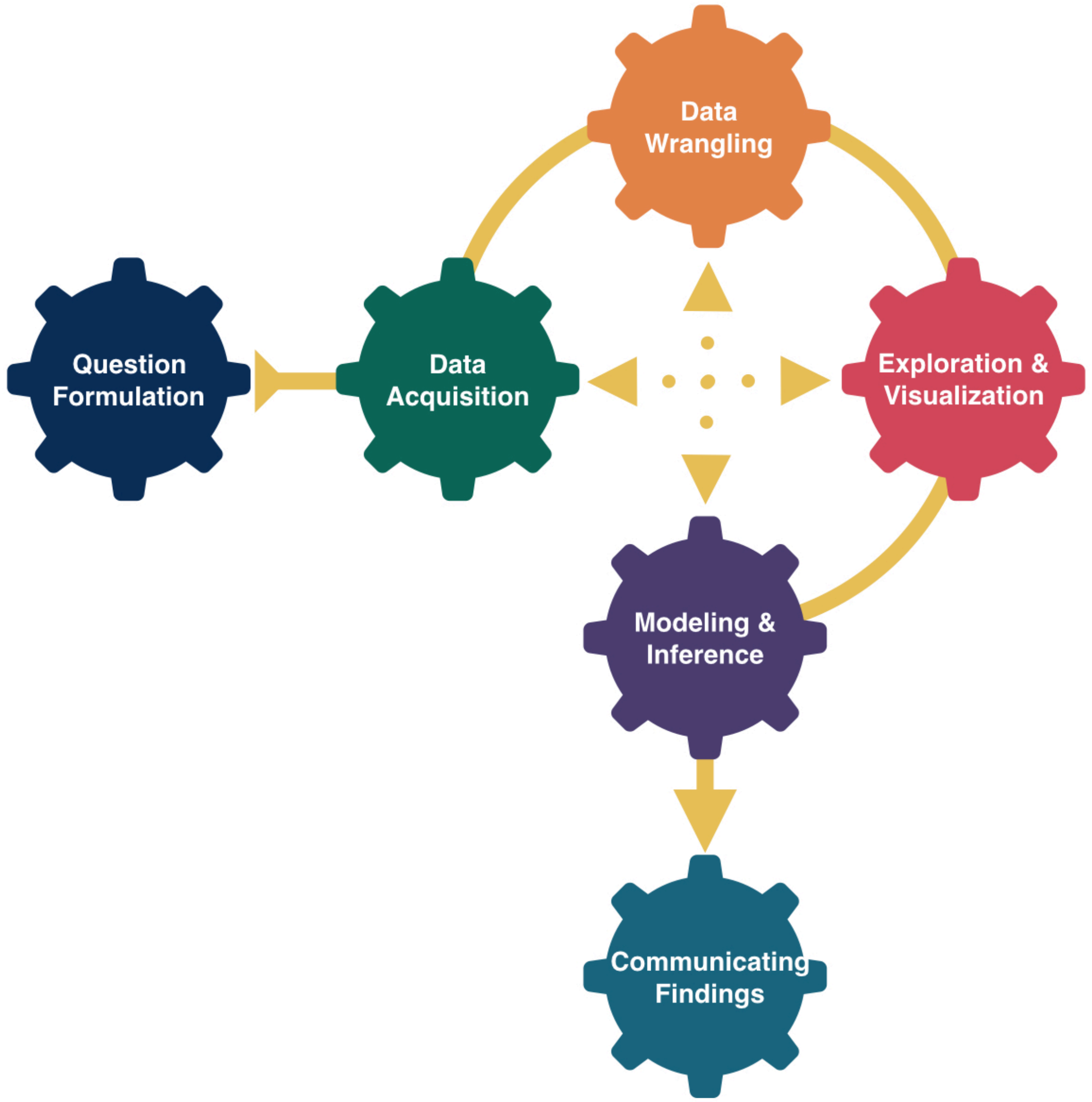


Decisions, Decisions

Kelly McConville

DATA 100

Week 10 | Spring 2026

Announcements

- 🎉We are now accepting data science student fellow applications for DATA 100.
  - About 10-12 hours of work per week.
  - Primary responsibilities: Work on a collaborative, mentored data or AI project, attend relevant meetings, and participate in DCDS ambassador activities.

Goals for Today

- Coding goals (DATA 100 & beyond)
- Advice on the next data science class
- Hypothesis testing example
- Decisions in a hypothesis test
  - Types of errors
- The power of a hypothesis test

Now that you have discovered your love of data, what class should you take next?

That Next Data Science Class

- But first… Common Question: How should I describe my post-DATA 100 coding abilities?
- Potential Answer: You have learned how to write code to analyze data. This includes visualization (`ggplot2`), data wrangling (`dplyr`), data importation (`readr`), modeling, inference (`infer`) and communication (with `Quarto`).
- Follow-up Question: So what coding is there left to learn?
- Answer: Learning how to program. This includes topics such as control flow, iteration, creating functions, and vectorization. Also, learning how to use an AI tool effectively and ethically to support your code generation and how to debug AI code suggestions.

That Next Coding Class

- CSCI 187. Creative Computing & Society: Computing, Creativity & the Social Good.
- CSCI 203. Introduction to Computer Science.
- ANOP 203. Introduction to Programming for Business Analytics.

That Next Modeling Course

- STAT 217. Statistics II
- CSCI 349. Applied Machine Learning.
- ANOP 330. Predictive Analytics: Machine Learning Fundamentals for Business.
- Many of the upper-level stats courses are modeling courses (but they do have pre-reqs).

That Next Visualization Course

- STAT 230. Data Visualization & Computing
- ANOP 270. Data Visualization for Business Analytics.

Feel free to stop by office hours or send me a Slack note if you have advising questions!

Now: A closer look at the conclusions of a hypothesis test

Hypothesis Testing Framework

- Set-up hypotheses:
  - Be able to express in words and in terms of parameters.
- Collect data.
- Assume H_o is correct.
- Compute a p-value to quantify the likelihood of the sample results using a test statistic.
- Draw a conclusion about the alternative hypothesis.

Hypothesis Testing: Decisions, Decisions

Once you get to the end of a hypothesis test you make one of two decisions:

- P-value is small.
  - Reject H_o.
  - I have evidence for H_a.
- P-value is not small.
  - Fail to reject H_o.
  - I don't have evidence for H_a.

Sometimes we make the correct decision. Sometimes we make a mistake.

Hypothesis Testing: Decisions, Decisions

Let's create a table of potential outcomes.

α = prob of Type I error **under repeated sampling** = prob reject H_o when it is true

β = prob of Type II error **under repeated sampling** = prob fail to reject H_o when H_a is true.

Hypothesis Testing: Decisions, Decisions

Typically set α level beforehand.

Use α to determine "small" for a p-value.

- P-value ~~is small~~ $< \alpha$.
  - I have evidence for H_a. Reject H_o.
- P-value ~~is not small~~ $\geq \alpha$.
  - I don't have evidence for H_a. Fail to reject H_o.

Hypothesis Testing: Decisions, Decisions

Question: How do I select α?

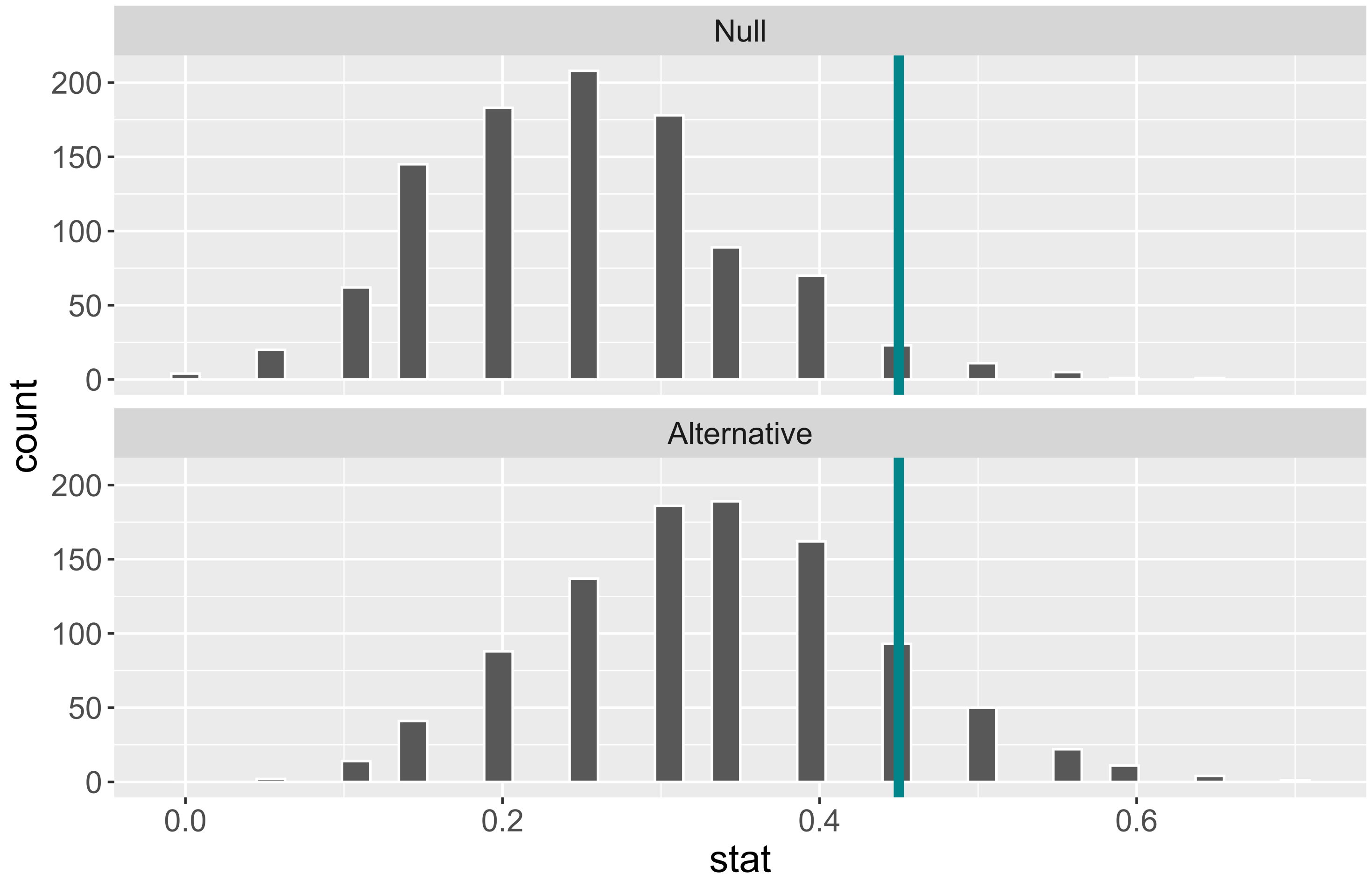
- Will depend on the convention in your field.
- Want a small α and a small β. But they are related.
  - The smaller α is the larger β will be.
- Choose a lower α (e.g., 0.01, 0.001) when the Type I error is worse and a higher α (e.g., 0.1) when the Type II error is worse.
- Can't easily compute β. Why?
- One more important term:
  - Power = probability reject H_o when the alternative is true.

Example

Suppose we have a baseball player who has been a 0.250 career hitter who suddenly improves to be a 0.333 hitter. He wants a raise but needs to convince his manager that he has genuinely improved. The manager offers to examine his performance in 20 at-bats.

Ho:

Ha:

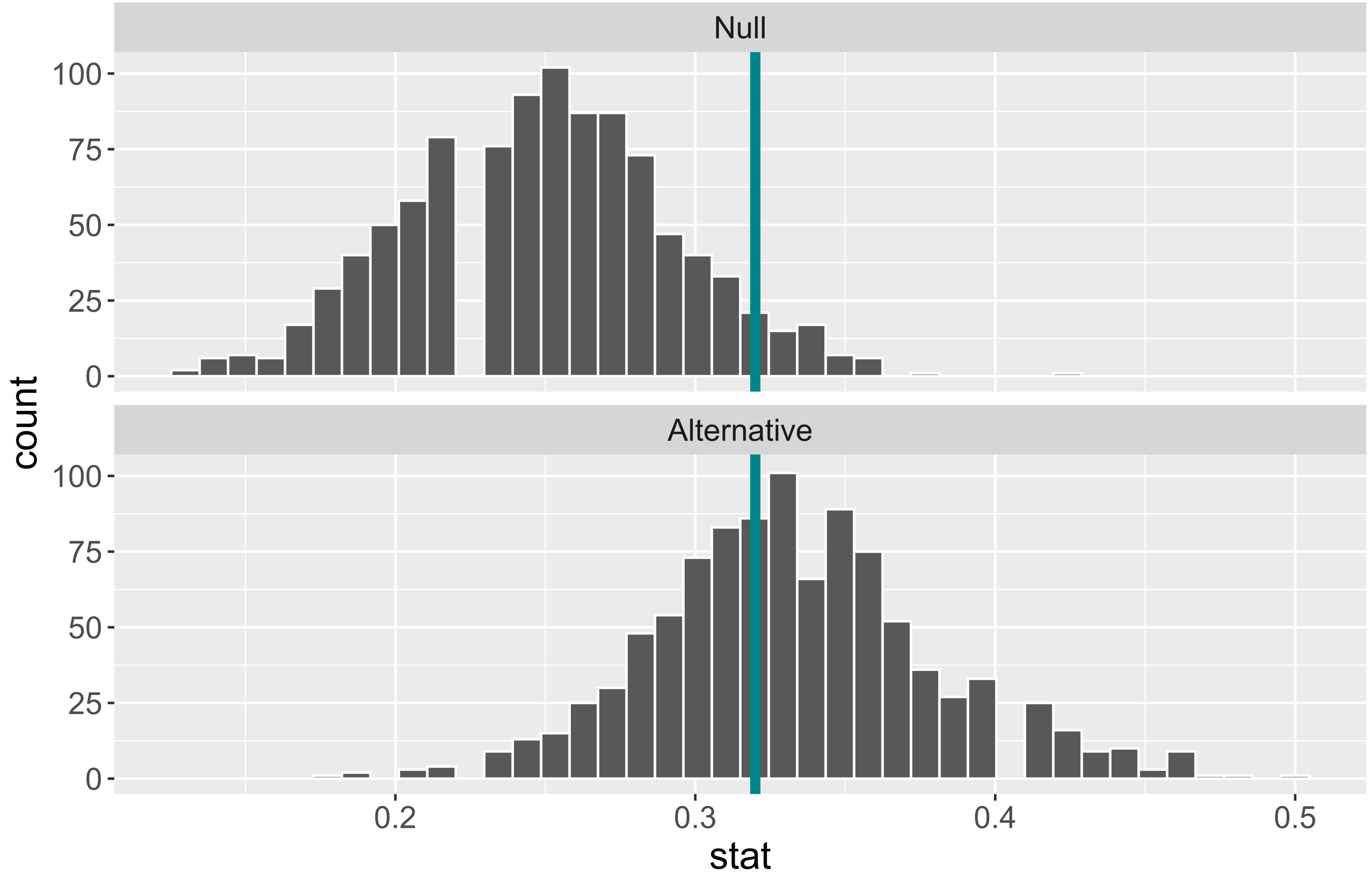


- When α is set to 0.05, he needs to hit 9 or more to get a small enough p-value to reject H_o.
- When α is set to 0.05, the power of this test is 0.181.
- Why is the power so low?
- What aspects of the test could the baseball player change to increase the power of the test?

Example

Suppose we have a baseball player who has been a 0.250 career hitter who suddenly improves to be a 0.333 hitter. He wants a raise but needs to convince his manager that he has genuinely improved. The manager offers to examine his performance in ~~20~~ 100 at-bats.

What will happen to the power of the test if we increase the sample size?

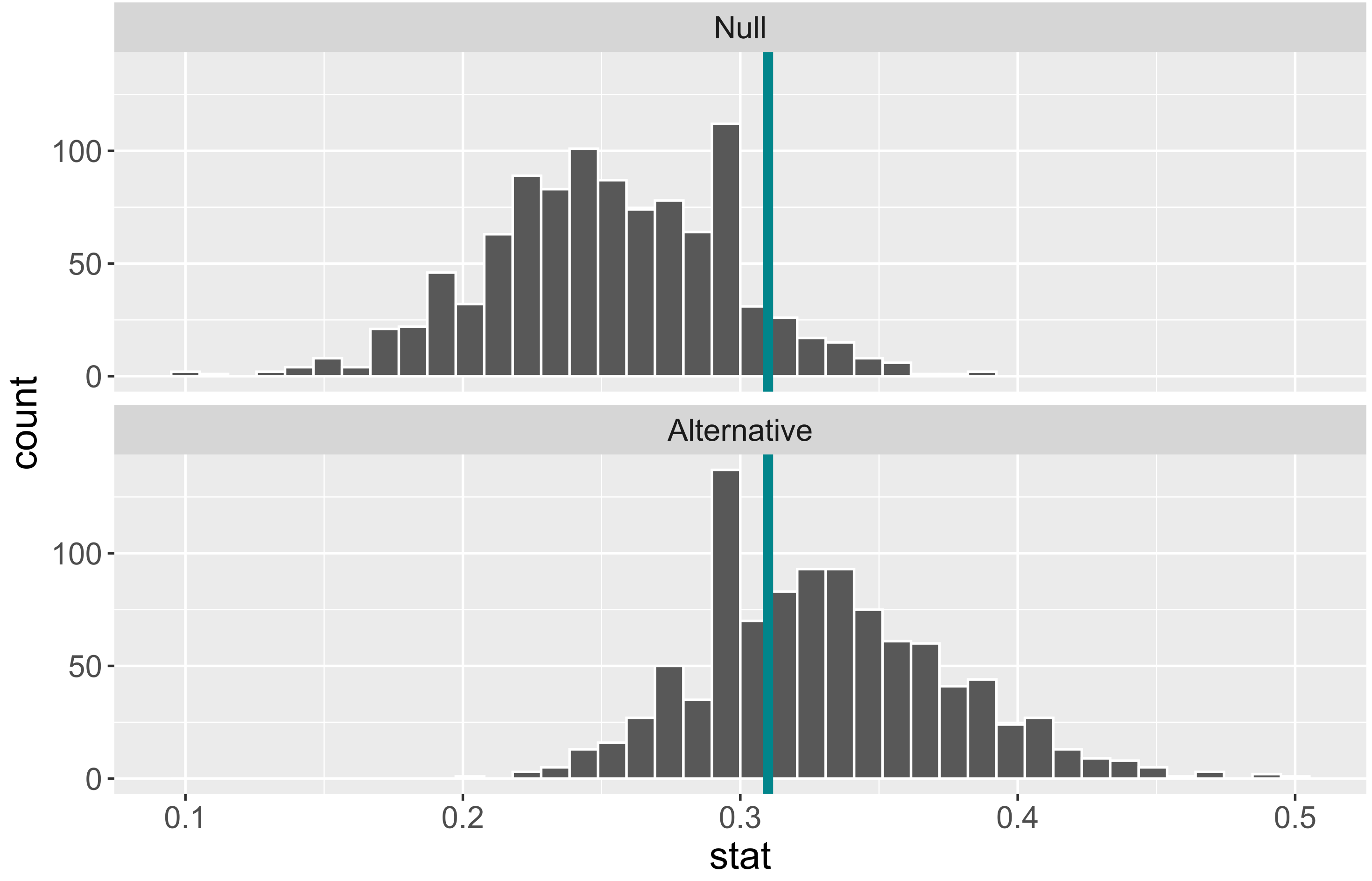


- Increasing the sample size increases the power.
- When α is set to 0.05 and the sample size is now 100, the power of this test is 0.55.

Example

Suppose we have a baseball player who has been a 0.250 career hitter who suddenly improves to be a 0.333 hitter. He wants a raise but needs to convince his manager that he has genuinely improved. The manager offers to examine his performance in ~~20~~ 100 at-bats.

What will happen to the power of the test if we increase α to 0.1?

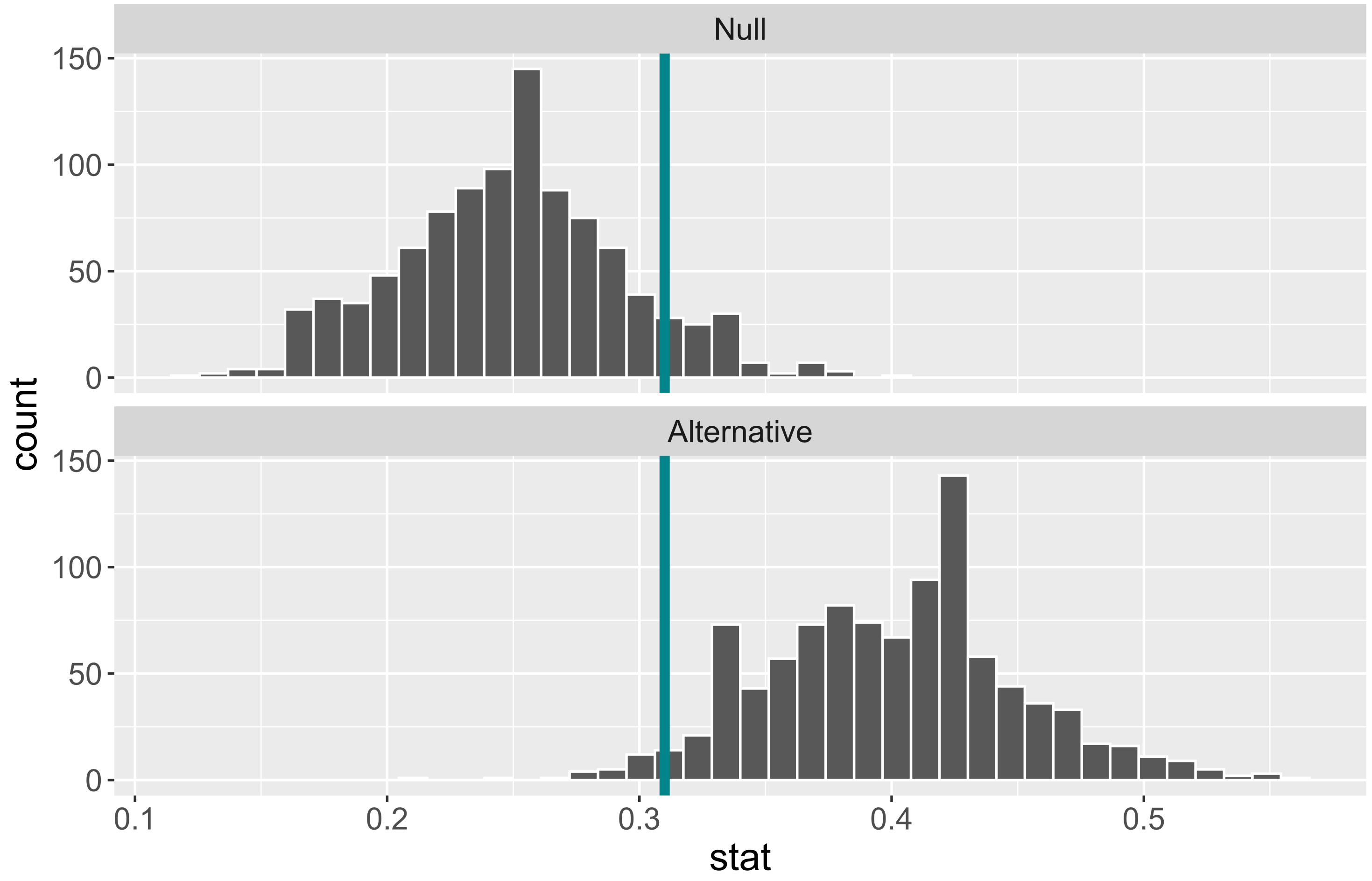


- Increasing α increases the power.
  - Decreases β.
- When α is set to 0.1 and the sample size is 100, the power of this test is 0.64.

Example

Suppose we have a baseball player who has been a 0.250 career hitter who suddenly improves to be a ~~0.333~~ 0.400 hitter. He wants a raise but needs to convince his manager that he has genuinely improved. The manager offers to examine his performance in ~~20~~ 100 at-bats.

What will happen to the power of the test if he is an even better player?



- **Effect size**: Difference between true value of the parameter and null value.
  - Often standardized.
- Increasing the effect size increases the power.
- When α is set to 0.1, the sample size is 100, and the true probability of hitting the ball is 0.4, the power of this test is 0.96.

Will talk about computing power later!

Thoughts on Power

- What aspects of the test did the player actually have control over?
- Why is it easier to set α than to set β or power?
- Considering power before collecting data is very important!
- The danger of under-powered studies
  - EX: Turning right at a red light

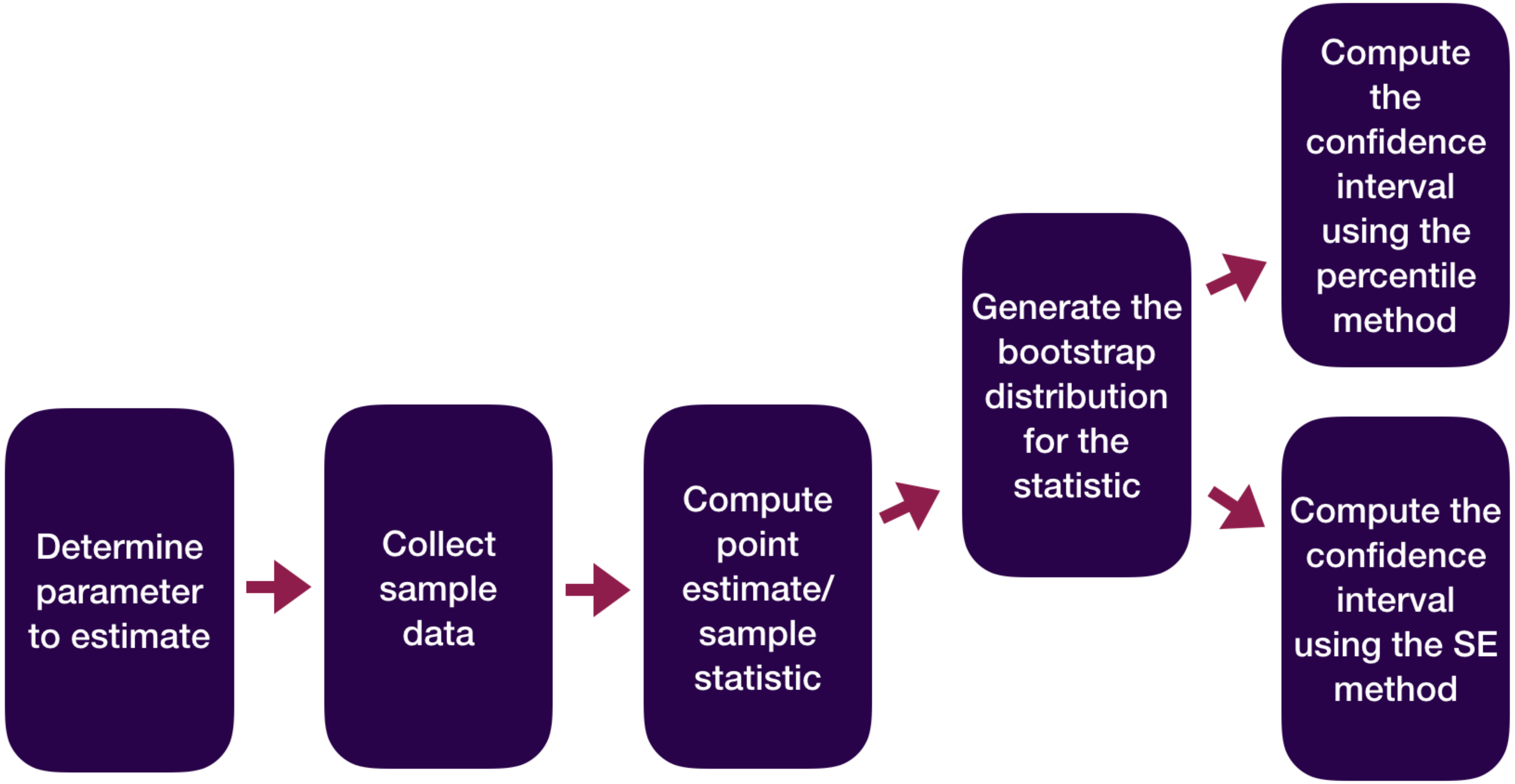
Reporting Results in Journal Articles

Results

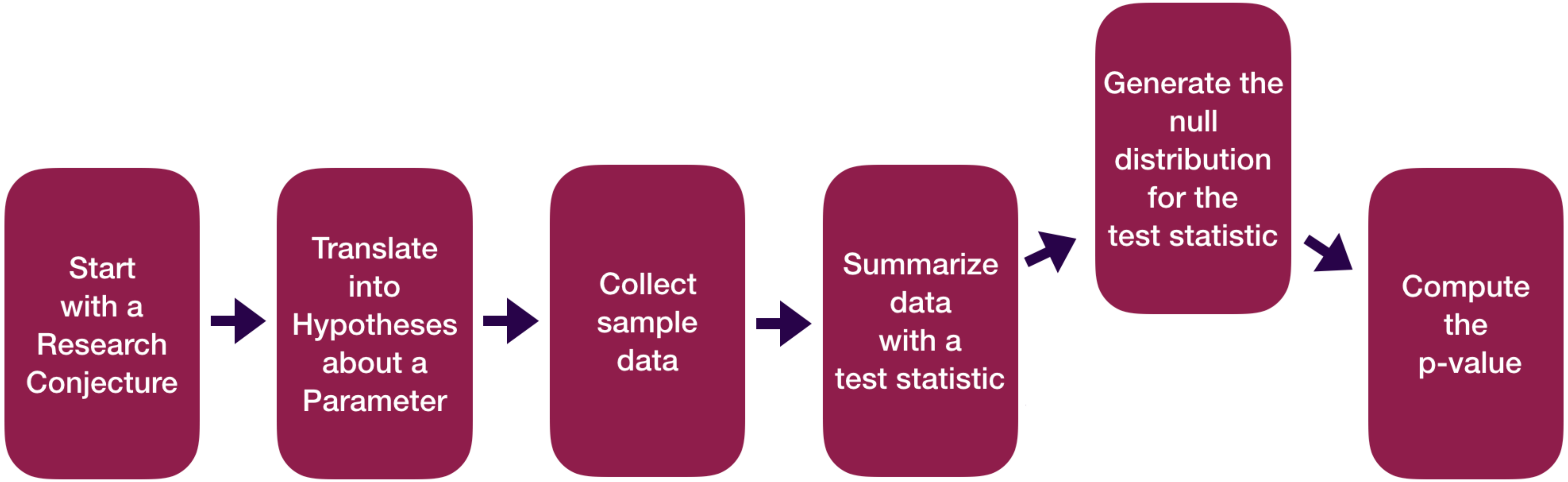
Overall hit rate. As in the earlier meta-analysis, receivers' ratings were analyzed by tallying the proportion of hits achieved and calculating the exact binomial probability for the observed number of hits compared with the chance expectation of .25. As noted earlier, 240 participants contributed 354 sessions. For reasons discussed later, Study 302 is analyzed separately, reducing the number of sessions in the primary analysis to 329.

As Table 1 shows, there were 106 hits in the 329 sessions, a hit rate of 32% (z = 2.89, p = .002, one-tailed), with a 95% confidence interval from 30% to 35%. This corresponds to an effect size (π) of .59, with a 95% confidence interval from .53 to .64.

Statistical Inference Zoom Out – Estimation

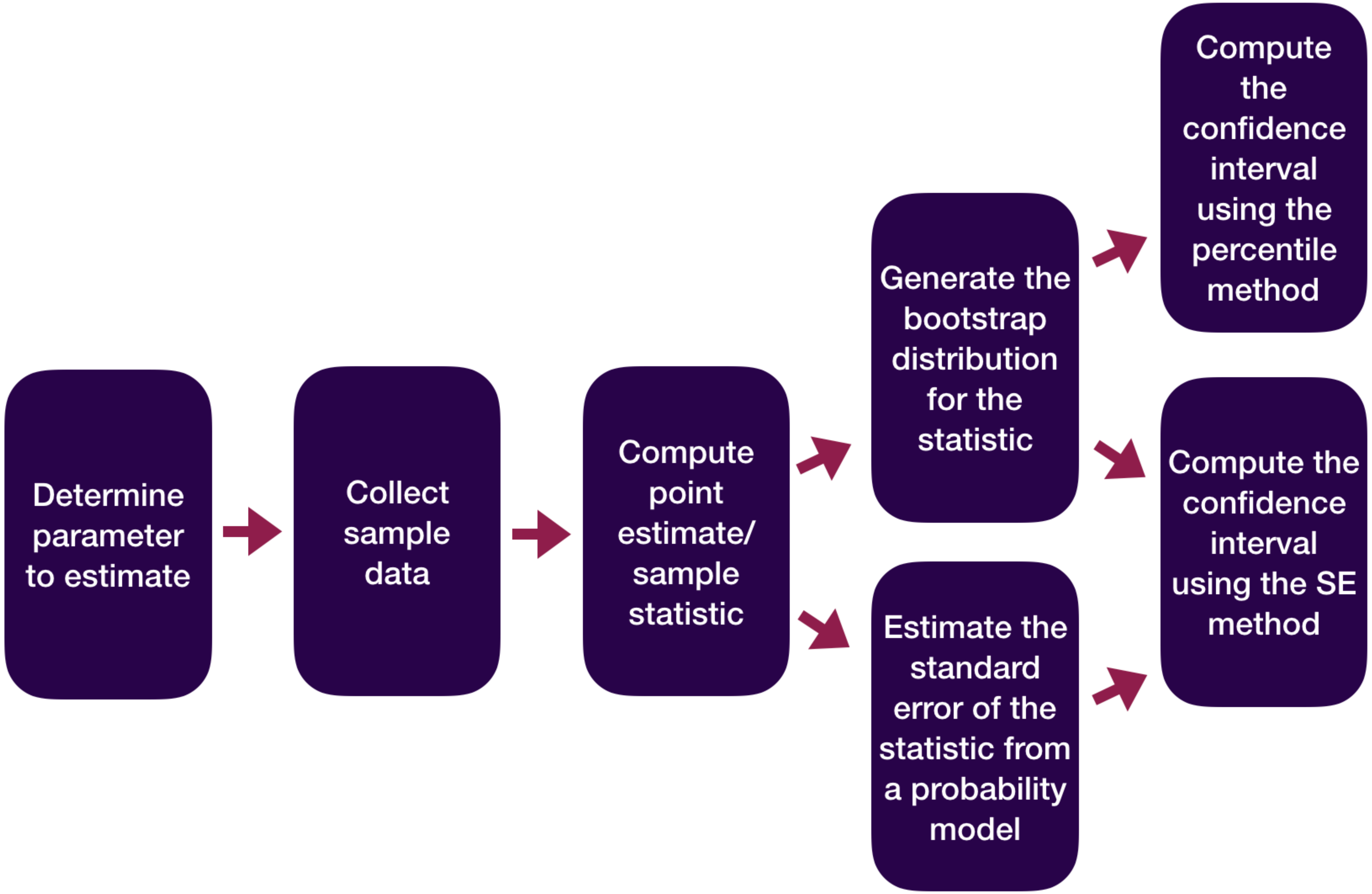


Statistical Inference Zoom Out – Testing

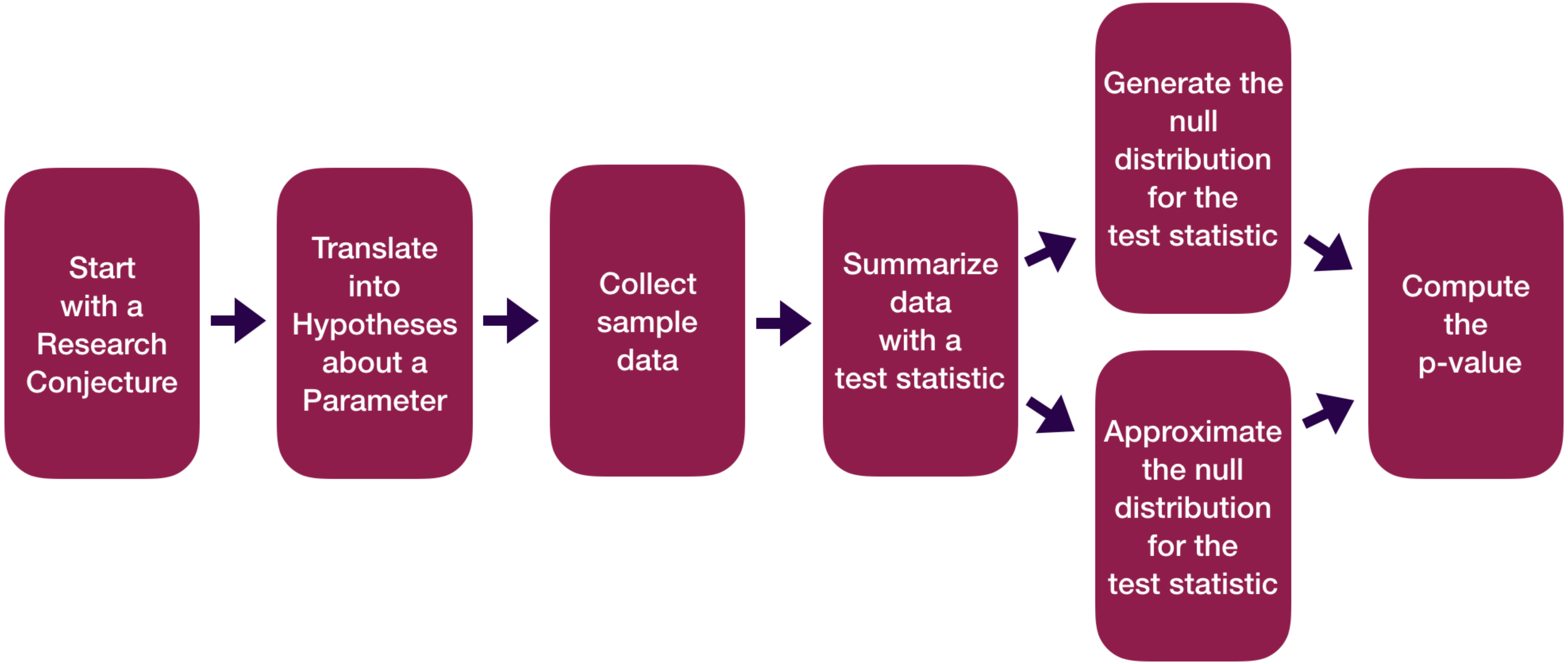


Question: How did folks do inference before computers?

Statistical Inference Zoom Out – Estimation



Statistical Inference Zoom Out – Testing



Question: How did folks do inference before computers?

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